

Personal Project by **Rangga Hadiwibowo**

Indonesia Healthcare Infrastructure Analysis



Hospital Data

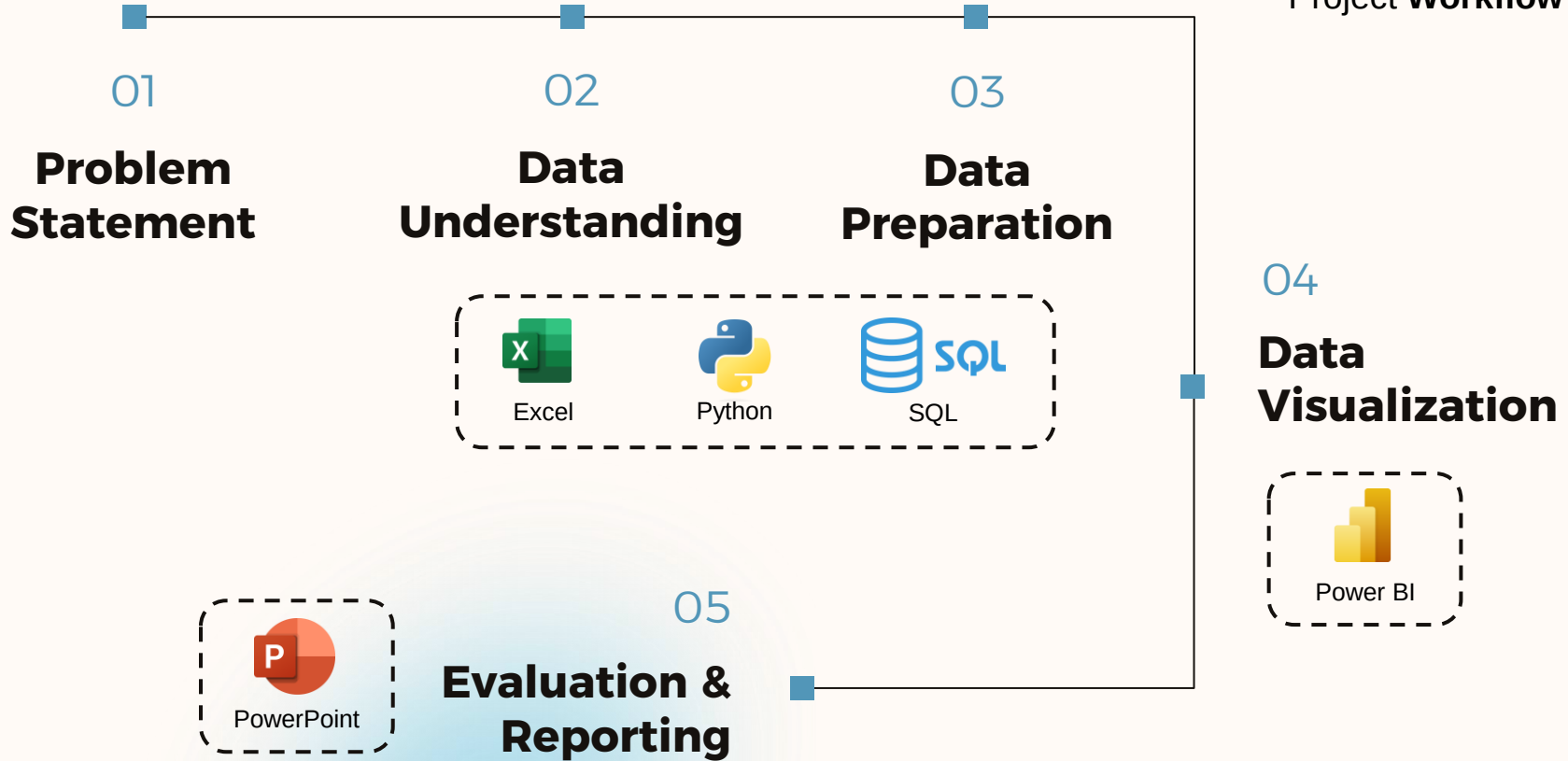
- **Original Title:** *Hospital Data in Indonesia*
- **Last Update:** February 1, 2025
- **Rows:** 3091
- **Columns:** 12



Population Data

- **Original Title:** *Population, Population Growth Rate, Percentage Distribution of Population, Population Density, and Population Sex Ratio by Province, 2026*
- **Last Update:** February 6, 2026
- **Rows:** 43
- **Columns:** 6

Project Workflow





01 General Overview of the Datasets

Inspect the structure of each dataset to identify column names, data types, and missing values. Based on this information, determine which columns will be used in the following steps. Column names and their order are also adjusted to improve readability.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3155 entries, 0 to 3154
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   id                    3155 non-null   int64
 1   nama                  3155 non-null   object
 2   propinsi              3155 non-null   object
 3   kab                   3155 non-null   object
 4   alamat                3155 non-null   object
 5   jenis                 3155 non-null   object
 6   kelas                 3155 non-null   object
 7   status_blu           3155 non-null   object
 8   kepemilikan           3155 non-null   object
 9   total_tempat_tidur    3155 non-null   int64
10   total_layanan         3155 non-null   int64
11   total_tenaga_kerja    3155 non-null   int64
dtypes: int64(4), object(8)
memory usage: 295.9+ KB
```

Findings

- All column in the hospitals dataset has no Null values.
- All column in the population dataset has Null values.
- All column in the hospitals dataset will be used for future steps.
- Only several column in the population dataset will be used:
 - 'province'
 - 'population_thousand'
 - 'population_density'



02 Detect Duplicate Values in Key Columns

The primary key in each dataset acts as the unique identifier for every row. This step checks whether duplicate values exist in the primary key fields.

Findings Key in both datasets has no duplicate values.

03 Detect Zero or Negative Numbers in Specific Columns

After reviewing the column descriptions, numerical values are inspected. The **'total_beds'**, **'total_labors'**, and **'total_services'** columns should not contain zero or negative values, so this step identifies any invalid records.

Findings There are zero or negative numbers in **'total_beds'** and **'total_labors'**, while **'total_services'** does not have it.



04 Detect Outlier Anomalies with Z-Score Method

Besides invalid values, large datasets may also contain outliers. This step uses the Z-score method to identify extreme values, while domain knowledge is used to determine whether the detected values should be treated as anomalies.

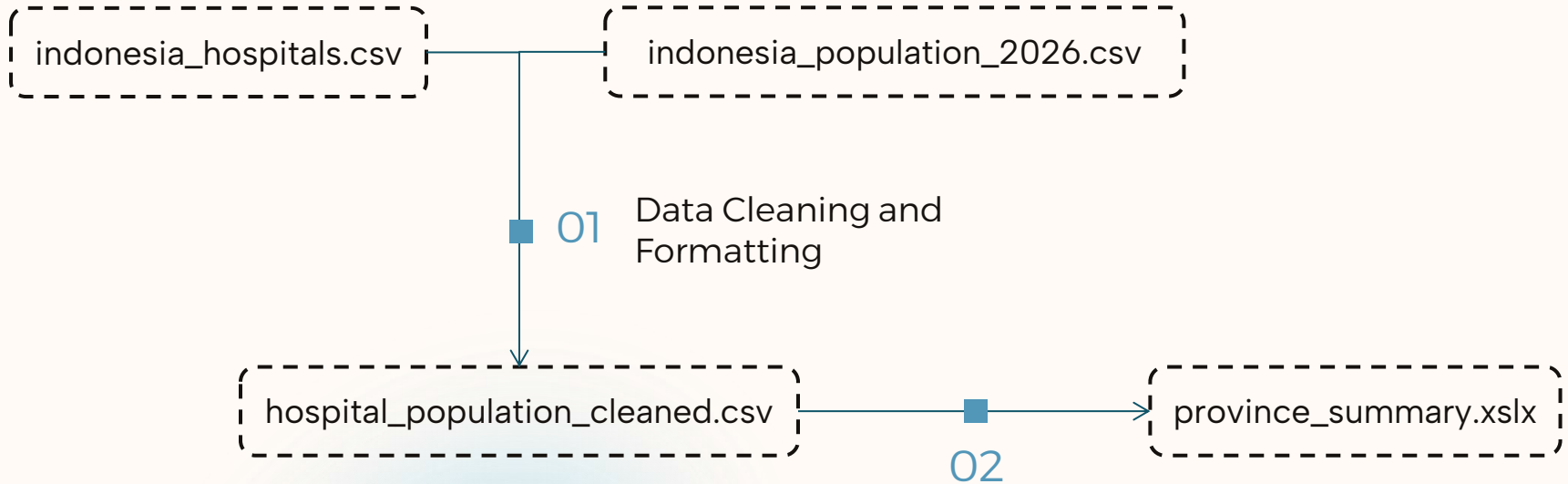
Findings There are 2 value considered as anomalies in **'total_beds'**, while **'total_services'** and **'total_labors'** does not have it.

05 Compare Shared Column between Datasets

The shared column is used as the join key between both datasets, so its values must be consistent. This step identifies any differences in the shared column before joining.

Findings From all of item in shared column of Province, only **"Yogyakarta"** has different value, which has **"DI Yogyakarta"** in the other dataset.

Data Preparation



Prepare Key Indicator using
Business Questions



01 Data Cleaning and Formatting

Perform data cleaning based on the findings from the Data Understanding phase. In addition, reformat the datasets to improve consistency and simplify further analysis.

- Handling null values
- Handling zero or negative numbers
- Handling outlier anomalies
- Standardize Province Names
- Merge datasets



02 Prepare Key Indicator using Business Questions

How many hospitals are there in each province?

Measures the distribution of healthcare facilities and identifies provinces with higher or lower hospital availability.

What is the number of bed per 1K population in each province?

Evaluates whether hospital bed capacity is sufficient relative to the size of the population.

How many beds are there in each province?

Indicates the healthcare system's capacity to accommodate patients in each province.

How many class A and B hospitals are there in each province?

Shows the distribution of high-tier referral hospitals capable of providing specialized and complex medical services.



Data Visualization

The cleaned and aggregated dataset was imported into Power BI to create an interactive dashboard. The dashboard enables users to quickly identify regional disparities in hospital availability, healthcare capacity, and access to advanced healthcare services.



Hospital Distribution Analysis

The World Health Organization (WHO) uses hospital beds per population as an international health system indicator to assess healthcare capacity. In this analysis, the metric is used to compare the availability of inpatient healthcare resources across Indonesian provinces and identify regional disparities.

Beds per 1K Population



Papua	3.2
Sulawesi Utara	2.7
DKI Jakarta	2.7
Yogyakarta	2.4
Kalimantan Utara	2.4
Bali	2.1
Papua Barat	2.0
Aceh	1.9
Gorontalo	1.9
Kepulauan Riau	1.9

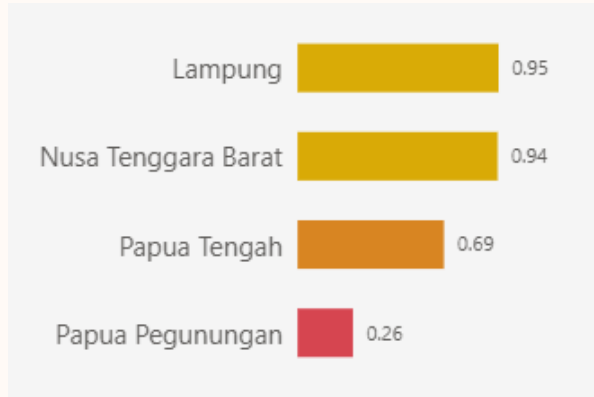
Data Evaluation

<1	Low bed availability; limited inpatient capacity. Common in many lower-resource health systems.
1-3	Moderate bed availability. Many developing and middle-income countries fall within this range.
3-8	Typical range for many developed countries.
>10	Very high bed availability, seen in countries such as Japan and South Korea.

Indonesia beds per 1K population index ranging from 0.3 to 3.2, with average of 1–2 in most of provinces. The top 5 highest beds per 1K population is **Papua, Sulawesi Utara, DKI Jakarta, Yogyakarta, and Kalimantan Utara**, with Papua is the only province within group of many developed countries had.

Hospital Distribution Analysis

Data Evaluation



Province	Population	Total Beds
Lampung	9.623.800	9.116
NTB	5.815.300	5.482
Papua Tengah	1.510.800	1.042
Papua Pegunungan	1.501.900	392

Even though most of the province across Indonesia has above WHO recommendation of beds per 1K beds index, there are **4 province that is still considered low in beds**, that is **Lampung, Nusa Tenggara Barat, Papua Tengah** and **Papua Pegunungan**.

High-Tier Hospital Distribution

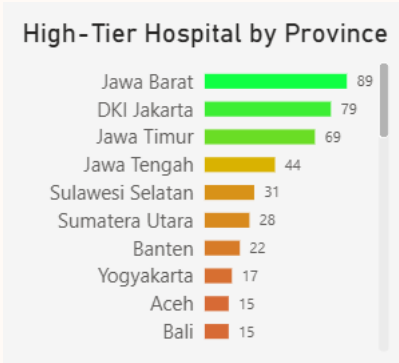
Data Evaluation

Class	Typical Services
Class A	Comprehensive specialist & subspecialist services, advanced diagnostics, complex surgery, ICU, cancer treatment, organ transplant support, etc.
Class B	Specialist and several subspecialist services, advanced surgery, intensive care, emergency referral.
Class C	Basic specialist services (internal medicine, surgery, pediatrics, obstetrics & gynecology).
Class D	General medical care, emergency care, stabilization, limited specialist services.

While the hospital beds per 1,000 population indicator suggests that many Indonesian provinces have established a reasonable level of healthcare capacity, this metric alone is insufficient to evaluate healthcare accessibility. **Access to advanced medical services depends not only on bed availability** but also on the presence of higher-tier hospitals capable of providing **specialized treatment, sophisticated medical equipment, and comprehensive referral services.**

High-Tier Hospital Distribution

Data Evaluation



In this analysis, high-tier hospitals refer to the combined number of Class A and Class B hospitals, which provide advanced medical services and specialized healthcare. The distribution of these hospitals is highly concentrated in a few provinces, particularly **West Java**, **DKI Jakarta**, and **East Java**. In contrast, many provinces have only a limited number of high-tier hospitals, while **Papua Barat**, **Papua Barat Daya**, **Papua Pegunungan**, and **Papua Tengah** have none.

Key Findings

Conclusion

- Healthcare infrastructure is available across all Indonesian provinces, with a total of 3,155 hospitals, 394,985 hospital beds, and 826,549 healthcare workers.

- Hospital beds per 1,000 population indicate that many provinces have established a reasonable level of inpatient healthcare capacity, suggesting that basic healthcare infrastructure is relatively widespread.

- However, the distribution of Class A and Class B hospitals remains highly concentrated, particularly in Java, while several provinces have few or no high-tier hospitals.

Recommendations

Although Indonesia has successfully expanded its basic healthcare infrastructure, equitable access to advanced healthcare services has not yet been achieved. The unequal distribution of high-tier hospitals suggests that patients in several provinces may face limited access to specialized services and rely on referrals to other regions. Therefore, **future healthcare planning should focus not only on increasing infrastructure but also on improving the equitable distribution of high-tier healthcare services.**

All files associated with this report can be accessed using the link below.

kaggle

[Kaggle Dataset](#)



[BPS Dataset](#)



[GitHub Files](#)